

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

BT2102-1 – BIOPROCESS PRINCIPLES AND CALCULATIONS (BT)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- An ideal gas mixture has the following composition (by volume %): $N_2 = 78$, $O_2 = 21$ & others = 1. What is the percent molar composition of oxygen gas in this mixture?
A) 78 B) 21
C) 26.9 D) 57
- Density of liquid water in terms of g/cc is
A) 1 B) 1000
C) 0.001 D) 995
- In a binary mixture, the mole fractions of compound A and compound B are equal to 0.5. The molecular weights of A = 100 and B = 50. The average molecular weight of the mixture is:
A) 100 B) 50
C) 75 D) 25
- Which of the following has a higher concentration?
A) 1 g/L B) 1 mg/L
C) 1 PPM D) 1 PPB
- Coefficient of performance (COP) is defined as
A) work input / latent heat of condensation B) work input / heat leakage
C) heat transferred / work input D) latent heat of condensation / work input
- What is the change in internal energy for cyclic process?
A) Zero B) A finite value
C) Q D) W
- Which of the following is a path function?
A) Internal Energy B) Work
C) Enthalpy D) Entropy
- The specific heat (C_p) of an ideal gas is 28.316 J/mol.K. What is the value of C_v for this gas?
A) 28.316 B) 20
C) 1.41 D) Data insufficient
- In a drying operation, the solids fraction in the water vapor stream is:
A) 1.0 B) Zero
C) Any value between 0 and 1 D) 0.5
- Leaching operation can be termed as:
A) Solid-Liquid operation B) Liquid-Liquid operation
C) Gas-Liquid operation D) Gas-Liquid-Solid operation
- While solving the material balance equations for Crystallization process, how do you estimate the concentration of salt in the liquid solution of the product stream?
A) Using water of hydration data B) Using solubility data
C) Using humidity chart D) Using specific heat data
- In extraction operation, the two solvents must be
A) Immiscible with each other B) Completely miscible with each other
C) Equally miscible with the solute D) Immiscible with solute molecule

13. Which of these is best example for energy transfer process without reaction?
 A) ATP hydrolysis B) Any exothermic reaction
 C) Sterilization of fermentation fluid D) Incineration of biomass
14. In DPHE, the hot fluid enters the annulus and cold fluid enters the pipe from opposite direction. The type of flow in DPHE can be termed as:
 A) Parallel flow B) Counter current flow
 C) Mixed flow D) Cross flow
15. Kopp's rule and Dulong-Petit's rule can be used to predict specific heats of
 A) Solids only B) Solids and liquids
 C) Gases only D) Gases and Liquids
16. While performing energy balance in heat exchangers under steady state operation, which law of thermodynamics is applied?
 A) Zeroth Law B) First Law
 C) Second Law D) Third Law
17. If 1 mole of CH_4 is fed along with air and the reaction is incomplete combustion. The product stream contains which of these molecules?
 A) CH_4 only B) CH_4 , CO_2 and H_2O only
 C) CO_2 and H_2O D) CH_4 , CO_2 , H_2O , N_2 , and O_2 .
18. Degrees of reduction for NH_3 is
 A) zero B) -3
 C) 2 D) 2.6
19. Respiratory quotient in an elemental balance gives:
 A) Quantity of air consumed B) Yield of CO_2 per mole of O_2
 C) Quantity of biomass produced D) Yield of biomass per mole of O_2
20. The yield termed as Y_{X/O_2} represents
 A) Biomass yield per mole of carbon B) Product yield per mole of oxygen
 C) Biomass yield per mole of oxygen D) Product yield per mole of carbon

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) A solution is prepared by dissolving a certain quantity of a solid or liquid in a solvent. Explain how concentration can be explained in different methods. Also, give one suitable example for each.	8	L*2	1	1,2
	b) Define with a suitable example for the following: i) Thermodynamic system and its types, ii) Thermodynamic functions and iii) Equations of state for ideal and real gases.	8	L2	2	1
2.	a) CO_2 gas has a volume of 12 L at 125 kPa pressure and at a temperature of 350 K. If the pressure of the gas rises to 200 kPa, its volume decreases by 4 L, then what calculate the rise/drop in temperature?	8	L4	1	2
	b) Draw the PV curves for the following thermodynamic processes involving ideal gas: Isobaric process, Isochoric process, Adiabatic process, Isothermal process. Show the work done in each case.	8	L2	2	2
3.	a) Calculate the compressibility factor & molar volume for methanol vapor at 500 K & 10 bar using the truncated Virial equation. Virial coefficients are: $2.19 \times 10^{-4} \text{ m}^3/\text{mol}$ and $-1.73 \times 10^{-8} \text{ m}^6/\text{mol}^2$.	8	L3	1	1,2
	b) Give Kelvin Plank and Clausius statements of 2 nd law of thermodynamics. How is Carnot's cycle related to these statements? Relate the two concepts.	8	L3	2	1,2
Unit – II					
4.	a) One hundred kilograms of a dilute waste acid containing 30% sulfuric acid is to be fortified to 50% sulfuric acid using concentrated sulfuric acid of strength 96%. How many kilogram of concentrated sulfuric acid is required for this process?	8	L3	3	2,3

- b) It is desired to make a 34% by weight of caustic soda solution. It is done in two steps. In the first step caustic soda is dissolved in a dissolution tank in a correct quantity of water to produce a 55% solution. After the dissolution step the solution is taken to a dilution tank where some water is added to produce 34% by weight of salt. Calculate the weight ratio of water added in both the tanks. 8 L4 4 2,3
5. a) It is required to make 1000 kg of mixed acid containing 60% H_2SO_4 , 32% HNO_3 and 8% water by blending (i) the spent acid containing 11.3% HNO_3 , 44.4% H_2SO_4 and 44.3% H_2O , (ii) aqueous 90% HNO_3 , and (iii) aqueous 98% H_2SO_4 . All percentages are by mass. Calculate the quantities of the three acids required. 8 L3 3 2,3
- b) A stream of N_2 gas flowing at the rate of 10 kmol/h is heated from 303 K to 373 K. Determine the heat to be transferred to the gas stream. Specific heat data:
 $\text{CP (kJ/kmol.K)} = 29.59 - 5.14 \times 10^{-3} T + 11.18 \times 10^{-6} T^2 - 4.97 \times 10^{-9} T^3$ 8 L4 4 2
6. a) A polar drug is to be extracted from aqueous solution using pure ethyl acetate as solvent. The aqueous feed having 0.1% drug is fed continuously at the rate of 250 kg/h. The solvent is fed at the rate of 60 kg/h. The raffinate phase flows at the rate of 210 kg/h with drug concentration of 0.01%. Calculate extract phase flowrate and % recovery of drug. Note: All % are on w/w basis. 8 L4 3 2,3
- b) Write the energy balance equation giving suitable explanations and assumptions for: Fermenter and heat exchanger. 8 L3 4 2,3
- Unit – III**
7. a) A feed containing 60 mole% 'A', 30 mole% 'B' and 10 mole% inerts enter a reactor. 80% of 'A' reacts according to following stoichiometry: $2\text{A} + \text{B} \rightarrow \text{C}$. Find the composition of product stream on molar basis. 8 L3 5 2,3
- b) Production of single-cell protein from hexadecane is described by the following reaction equation:
 $\text{C}_{16}\text{H}_{34} + a \text{O}_2 + b \text{NH}_3 \rightarrow c \text{CH}_{1.66}\text{O}_{0.27}\text{N}_{0.20} + d \text{CO}_2 + e \text{H}_2\text{O}$
 Where $\text{CH}_{1.66}\text{O}_{0.27}\text{N}_{0.20}$ represents biomass. If $\text{RQ} = 0.43$, determine the stoichiometric coefficients. 8 L3 5 2,3
8. a) Calculate the standard heat of formation of benzoic acid using following data:
 Std. heat of formation of $\text{CO}_2 = -393 \text{ kJ/mol}$
 Std. heat of formation of $\text{H}_2\text{O} = -285 \text{ kJ/mol}$
 Std. heat of combustion of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$) = -3226 kJ/mol 8 L3 5 2,3
- b) Explain Proximate analysis and Ultimate analysis of fuels. 8 L2 5 2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
